1. Where Does Relation Extraction Get Its Evidence? A Context and Entity Ablation Study on TACRED

Problem introduction

Relation extraction aims to identify semantic relationships between entities mentioned in text. For example, given the sentence *"John joined Google in 2018,"* a system should recognize that **John** and **Google** participate in an employee of relationship. This task is important for information extraction, knowledge-base construction, and question answering.

A widely used benchmark for relation extraction is **TACRED (TAC Relation Extraction Dataset)**. TACRED contains more than 100,000 examples constructed from real-world text, with annotated subject–object entity pairs and relations such as per:employee_of, per:city_of_birth, and org:top_members/employees. Importantly, many examples require the model to use linguistic context beyond the two entity mentions themselves.

However, modern pretrained language models can achieve strong performance on TACRED without making it clear **what evidence they actually use to make their predictions**. A model may rely on the entities themselves, nearby lexical cues, entity types, or information occurring far away from the target entities. Understanding this behavior is important both for interpreting relation-extraction systems and for understanding what makes TACRED difficult.

Research question

This paper asks:

How much and what kind of textual context does a relation-extraction model actually need to predict relations in TACRED?

2. Are Relation Classifiers Learning Relations or Shortcuts?

Problem

SemEval-2010 Task 8 is a standard benchmark for classifying the semantic relation between two marked entities. It contains nine relation types plus Other, including relations such as Cause-Effect, Component-Whole, Instrument-Agency, and Member-Collection.

A challenge with benchmark datasets is that high-performing neural models may not necessarily learn the intended semantic relationship. Instead, they can exploit **surface-level shortcuts**, for example, particular words, entity types, syntactic patterns, or even the order in which entities appear.

This raises an important question:

When a model predicts a relation in SemEval-2010 Task 8, is it actually identifying the semantic relationship, or is it relying on shortcuts associated with particular relations?

Proposed study

Train a strong baseline such as RoBERTa or DeBERTa and then systematically test whether its predictions survive different forms of **counterfactual perturbation**.

For example:

1. Entity replacement

Original:

"The **doctor** used a **stethoscope**."

Replace entities:

"The **teacher** used a **stethoscope**."

Does the predicted relation remain Instrument-Agency?

2. Lexical perturbation

Replace non-critical words while preserving the sentence structure.

3. Entity-type perturbation

Replace entities with different entities having the same broad semantic type.

4. Relation-direction perturbation

Swap the subject and object and test whether the model correctly changes the directional relation.

5. Template-based examples

Construct controlled sentences where the same lexical patterns occur with different relations.

Main research question

Which superficial features do neural relation classifiers rely on, and how much does their performance change when those features are controlled or altered?

3. From Healthy Aging to Alzheimer's Disease: Evaluating Neural NLP Models for Detecting the Cognitive-Linguistic Continuum

1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder in which cognitive and language abilities can change over time. Language alterations may appear relatively early and can affect vocabulary, fluency, lexical retrieval, discourse organization, and semantic content. This has motivated growing interest in using speech and language as a non-invasive source of information for detecting cognitive impairment.

Natural language processing provides a way to automatically quantify these changes. Previous studies have used handcrafted linguistic features, word embeddings, recurrent neural networks, and transformer-based language models to distinguish cognitively impaired individuals from healthy controls. However, an important limitation is that many studies formulate the problem as a simple **binary classification task**: healthy versus AD.

That framing potentially misses an important clinical distinction: **mild cognitive impairment (MCI) represents an earlier stage of cognitive decline**, and detecting subtle linguistic differences associated with MCI may be more informative than simply detecting established dementia.

The **DementiaBank Delaware corpus** provides a useful resource for studying this problem. It contains speech and language samples from cognitively unimpaired adults and adults with MCI, collected using a standardized discourse protocol involving picture description, story narrative, procedural discourse, and personal narrative tasks.

The **DementiaBank Pitt corpus**, meanwhile, provides a complementary source for studying Alzheimer's/dementia-related language. It contains recordings from controls and individuals with dementia and has been used extensively for computational detection of AD-related language differences.

This creates an opportunity to study not merely whether a model can classify disease status, but whether it learns **generalizable linguistic signals associated with increasing cognitive impairment**.

2. Central research question

The paper asks:

Can neural language models learn linguistic representations that distinguish cognitively unimpaired individuals, people with MCI, and people with Alzheimer's-

related dementia, and do these representations generalize across datasets and discourse tasks?

This leads to three specific research questions.

RQ1 — Model comparison

How do different representations perform for detecting cognitive impairment?

Compare:

- Linguistic features + logistic regression/SVM
- Word embeddings + BiLSTM
- BERT/RoBERTa
- A lightweight transformer such as DistilBERT

The point is not simply to find the highest F1, but to see **how the representation affects the ability to detect subtle impairment**.

RQ2 — MCI versus AD

Do models learn a meaningful **severity gradient**?

Instead of only:

Control vs. AD

evaluate:

Control → MCI → AD

For example, train a three-class classifier:

CU | MCI | AD

and also perform pairwise experiments:

- CU vs. MCI
- MCI vs. AD
- CU vs. AD

The most interesting comparison is likely to be **CU vs. MCI**, because this tests whether models can detect more subtle linguistic changes.

RQ3 — Generalization

Do models learn a general cognitive-linguistic signal, or do they learn characteristics specific to a particular corpus or task?

For example:

Train: Delaware MCI vs. CU
Test: another task within Delaware

and, where labels/tasks permit:

Train: Pitt AD vs. control
Test: another AD-related benchmark.

This is important because recent work has already evaluated LLM adaptation on both ADReSSo/Pitt-derived data and Delaware MCI data, making simple "which transformer gets the best F1?" comparisons increasingly difficult to claim as novel.

3. The key novelty: a cognitive-impairment continuum

I would make this the **central contribution** of the paper.

Rather than treating the datasets as two unrelated classification problems, conceptualize them as:

Cognitively unimpaired → MCI → AD

Then ask whether the learned representations reflect this ordering.

For example, take the embedding produced by a model and visualize participants using PCA/UMAP:

```
    MCI
   /   \
  CU    AD
```

You can test whether representations become progressively more separated as cognitive impairment increases.

Even more interestingly, train a model to predict an **ordinal severity score**:

- CU = 0
- MCI = 1
- AD = 2

and compare it against ordinary multiclass classification.

The hypothesis would be:

If language models capture general cognitive-linguistic deterioration, their representations should exhibit an approximately ordered structure from CU to MCI to AD.

You don't have to claim that AD is literally a single linear progression—clinically, that would be too strong. Rather, you're testing whether **linguistic representations contain an ordered signal associated with diagnostic group**.

4. Ontology-Grounded LLMs for Detecting Breast Cancer Misinformation in Patient Conversations

1. Introduction and Motivation

Patients increasingly use online communities and large language models (LLMs) to seek information about cancer diagnosis, treatment, side effects, recurrence, and prognosis. While these resources can provide valuable support, patient-generated discussions also contain misconceptions and unsupported medical claims. Detecting such misinformation is challenging because it is often expressed indirectly, embedded in personal experiences, or formulated as a question containing an incorrect assumption.

Recent work demonstrates that this is a significant weakness of current LLMs. The **Cancer-Myth** benchmark contains 585 cancer-related patient questions with false presuppositions and found that even frontier LLMs frequently fail to identify and correct the underlying misconception. The benchmark also demonstrates an important problem with simple precautionary prompting: making models more cautious can increase false-positive corrections on questions that do not actually contain misinformation.

Breast cancer is an especially suitable domain for studying this problem. There is a large body of English-language patient-generated text from online breast cancer communities. Previous research has analyzed hundreds of thousands of breast-cancer-related posts from Breastcancer.org using NLP, demonstrating that these discussions contain rich information about treatments, symptoms, side effects, and patient experiences.

This project proposes to combine **domain-adapted LLMs, a purpose-built breast-cancer ontology, and evidence retrieval** to detect misinformation in patient conversations. Rather than asking an LLM simply whether a statement is true or false, the system will first convert natural-language statements into structured medical claims and then ground those claims in an explicit representation of breast-cancer concepts and relationships.

The central hypothesis is:

Explicit ontology grounding can make LLM-based misinformation detection more accurate, robust, and explainable than relying on the LLM's parametric knowledge or retrieval alone.

2. Research Problem

Consider the following patient statement:

“My tumor is HER2-positive, so chemotherapy won't work for me and I need HER2-targeted therapy instead.”

A conventional LLM may generate a plausible response based on its learned medical knowledge. However, the statement contains several distinct claims:

tumor

└── has_biomarker → HER2-positive

HER2-positive

└── associated_with → HER2-targeted therapy

HER2-positive

└── incompatible_with → chemotherapy

The final claim may be misleading because HER2-targeted therapy and chemotherapy can be used together depending on the clinical context.

The proposed system therefore treats misinformation detection as **structured claim verification**, rather than simple text classification.

3. Research Questions

RQ1: Domain adaptation

Does fine-tuning an LLM on breast-cancer-related language improve its ability to extract medical concepts, relations, and claims from patient-generated text?

RQ2: Ontology grounding

Does explicitly representing breast-cancer concepts and relationships improve misinformation detection compared with an LLM alone?

RQ3: Ontology versus RAG

Does an ontology provide information that ordinary retrieval-augmented generation (RAG) does not?

RQ4: Robustness

Does ontology grounding make models more robust to paraphrases, implicit misinformation, and counterfactual claims?

RQ5: Explanation faithfulness

Does grounding predictions in structured medical knowledge produce explanations that are more consistent with the evidence supporting the final decision?

4. Proposed Breast Cancer Ontology

The project will construct a compact **Breast Cancer Misinformation Ontology (BCMO)** rather than attempting to model all of oncology.

The ontology will be developed using established biomedical terminology as a starting point and refined specifically for the misinformation task.

Main concept classes

- Breast cancer
- Breast cancer subtypes
- Stage and grade
- Symptoms
- Risk factors
- Genetic factors
- Biomarkers
- Diagnostic tests
- Treatments
- Medications
- Procedures
- Side effects
- Recurrence
- Metastasis
- Prognosis
- Prevention and screening

For example:

Breast Cancer

├— Invasive Ductal Carcinoma

├— Invasive Lobular Carcinoma

├── DCIS

└── Triple-Negative Breast Cancer

Biomarker

├── ER

├── PR

└── HER2

Treatment

├── Surgery

├── Chemotherapy

├── Radiation

├── Endocrine Therapy

└── Targeted Therapy

The ontology will also define relations relevant to misinformation:

causes

increases_risk_of

decreases_risk_of

treats

prevents

detects

associated_with

has_side_effect

contraindicated_for

reduces_recurrence_risk

increases_recurrence_risk

improves_survival

An important feature will be the representation of **negation, uncertainty, and relation strength**. For example, the ontology should distinguish:

“Treatment X can reduce recurrence risk”

from:

“Treatment X always prevents recurrence.”

This is particularly important because medical misinformation often results from turning a probabilistic or conditional medical statement into an absolute claim.

5. Data Resources

The project will combine several English-language resources with different purposes.

5.1 Breast cancer patient language

Breastcancer.org provides a large source of patient-generated discussions. Previous research analyzed a sample of 500,000 posts and retained 362,074 posts concerning early breast cancer for NLP analysis of treatment experiences and symptoms.

This resource can provide the **domain language needed for LLM adaptation**, subject to appropriate data-access and licensing requirements.

Other research has also analyzed large-scale communication in the Breastcancer.org community, including 28,139 original posts and 663,748 replies.

5.2 MedQuAD

MedQuAD contains 47,457 medical question-answer pairs collected from 12 NIH websites, including Cancer.gov. The dataset contains useful metadata such as question type, question focus, UMLS concept identifiers, and semantic types.

The cancer-specific portion can be used for:

- domain adaptation,
- medical question understanding,
- concept normalization,
- ontology construction,
- and supervised fine-tuning.

5.3 Cancer-Myth

Cancer-Myth will serve primarily as an **adversarial evaluation benchmark**, rather than as the main fine-tuning dataset.

It contains 585 expert-verified cancer questions with false presuppositions, plus a 150-question non-false-presupposition set designed to measure whether systems over-correct benign questions.

This is particularly useful because the goal is not merely to detect explicit false statements, but to recognize **misinformation embedded in patient questions**.

5.4 Expert-annotated cancer misinformation

A recent dataset provides expert-annotated Reddit discussions covering breast, lung, colon, and prostate cancer and characterizes misinformation along multiple dimensions rather than using only a binary true/false label.

The **breast-cancer subset** is particularly attractive as an evaluation resource because it provides naturally occurring patient/community language rather than synthetic questions.

5. Learning to Detect Scam Progression from Conversational Text

Goal. Instead of treating a scam as a binary classification problem, this project studies whether deep-learning models can recognize **how a scam evolves over multiple conversational turns** and provides an early warning before the scam reaches its final stage.

Motivation

Many online scams are not a single fraudulent message. They develop gradually: a scammer establishes contact, builds trust, introduces a problem or opportunity, applies psychological pressure, and eventually requests money or sensitive information. The recently introduced **PreScam** benchmark is particularly well suited to this problem: it contains **11,573 structured scam conversations derived from 177,989 reports across 20 scam categories**, with conversation stages, scammer psychological actions, and victim responses.

Research Question

Can deep-learning models identify the progression of a scam from early conversational turns and predict when the interaction is becoming high risk?

Proposed Approach

Compare several increasingly sophisticated models:

1. **TF-IDF + Logistic Regression** — traditional NLP baseline.
2. **BERT/RoBERTa** — supervised deep-learning baseline.
3. **Long-context Transformer** — incorporates multiple conversational turns.
4. **Fine-tuned LLM** — e.g., a smaller open-source LLM with LoRA.
5. **Progression-aware model** — explicitly represents the sequence of scam stages and psychological actions.

The primary task would be:

Given the conversation up to turn *t*, predict whether it is approaching the **Termination** stage.

A secondary, smaller experiment could predict the **next scammer action**, such as building trust, creating urgency, requesting information, or requesting payment. These are already supported by the structure of PreScam.

Key Contribution

Rather than asking:

“Is this message a scam?”

the paper asks:

“Is this conversation becoming a scam, and how early can we detect the escalation?”

This allows evaluation using **F1, AUROC/AUPRC, and early-detection/alert-time metrics**, making the work relevant to real-time scam prevention.

Expected Contribution

The paper would provide:

- a comparison of classical NLP, Transformers, and fine-tuned LLMs;
- an analysis of how much **conversation history** is necessary for reliable detection;
- an analysis of which **psychological techniques** are most predictive of escalation;
- an evaluation of the trade-off between **early detection and false alarms**.

6. Ideas of Survey Papers

i. How Reliable Are NLP Models for Dementia Detection? A Survey of Datasets, Evaluation, and Reproducibility
ii. Deep Learning for Detecting Deceptive Language in Online Communication: A Survey
iii. Ontology-Grounded Large Language Models for Misinformation Detection: A Survey of Methods and Applications
iv. NLP for Detecting Suspicious Communication in Enterprise Email: A Survey
v. LLMs for Clinical Decision Support in heart disease: A Survey of Evidence, Evaluation, and Safety

7. Semester NLP Research Paper Expectations

The goal of the semester project is to give students experience conducting a **small, complete NLP research study** that could potentially be developed into a workshop or student research publication. Students are not expected to develop a completely new deep-learning architecture. Instead, the emphasis should be on identifying a focused research question, designing a rigorous experiment, and producing meaningful analysis using existing datasets and modern NLP methods.

Expected Paper

Students should produce a **8 pages or more research paper, excluding references**, following the general structure of an NLP conference paper. The paper should contain:

1. **Introduction and Research Question**
   Clearly motivate the problem and formulate one main research question, supported by two or three secondary questions.

2. **Related Work**
   Review approximately **15–30 relevant research papers**, emphasizing previous datasets, NLP methods, and limitations in existing work.

3. **Dataset and Methodology**
   Use **one or two established datasets** whenever possible. Students should describe the dataset, preprocessing, experimental setup, and evaluation protocol in sufficient detail for reproduction.

4. **Model Comparison**
   Students should evaluate approximately **3–5 approaches**, including:

   - at least one traditional NLP/ML baseline, such as TF-IDF + Logistic Regression or SVM;
   - at least one neural/deep-learning model;
   - at least one Transformer or fine-tuned LLM approach.

5. **Research Experiment**
   The project should contain at least one experiment that goes beyond simply comparing models. Examples include studying the effect of conversation context, early detection, domain adaptation, ontology grounding, cross-dataset generalization, or different linguistic representations.

6. **Ablation and Error Analysis**
   Students should investigate *why* models succeed or fail. This can include an ablation study and qualitative analysis of approximately **30–50 incorrectly classified examples**.

7. **Results and Discussion**
   Report appropriate metrics such as **precision, recall, F1, and AUROC/AUPRC**, and discuss the strengths, weaknesses, and limitations of the approaches.

8. **Reproducibility**
   Students should provide their code, experimental configuration, and sufficient documentation to allow others to reproduce the main results.

Scope

The project is intended to be completed over approximately **12–15 weeks by a team of 2–3 students**. Students should not be expected to train an LLM from scratch, collect a massive new dataset, or develop a completely novel neural architecture. The research contribution can instead come from a **new comparison, evaluation, adaptation, analysis, or experimental question**.

The key distinction between a simple class project and a research paper is the **research question**. For example, *"We compare BERT and RoBERTa for dementia detection"* is primarily a benchmarking exercise. A stronger research question would be:

"How well do Transformer models generalize across independent datasets for dementia detection, and to what extent are previously reported results affected by dataset and evaluation methodology?"

Similarly, rather than simply *"using BERT to detect scams,"* students could investigate:

"How early can conversational Transformer models detect that an apparently normal interaction is progressing toward a scam?"

Thus, the primary objective is not necessarily to achieve state-of-the-art performance, but to conduct a **focused, reproducible, and scientifically meaningful investigation** that demonstrates students' ability to formulate an NLP research problem, work with real datasets, apply deep-learning methods, design-controlled experiments, and critically analyze the results.

Model paper: https://aclanthology.org/2023.acl-long.668.pdf

8. Semester NLP Survey Paper Expectations

The goal of the survey project is to give students experience conducting a **focused, critical literature review of an NLP research problem**. The paper should go beyond summarizing existing work. Students are expected to identify a well-defined research question, systematically organize the literature, compare datasets and methodologies, identify limitations and research gaps, and develop a clear perspective on future research directions.

Expected Paper

Students should produce a **10 pages or more survey paper, excluding references**, following the general structure of an NLP conference paper. The paper should contain:

1. **Introduction and Research Question**
   Clearly define the NLP application being surveyed and explain why it is important. The paper should have one central question, such as:
   - How reliable are deep-learning models for dementia detection?
   - How well do Transformer models generalize across datasets?
   - What datasets and evaluation practices are used for detecting scams?
   - How are LLMs being used for medical misinformation detection?
2. **Systematic Literature Search**
   Students should identify approximately **25–40 relevant research papers** using appropriate academic databases and search strategies. The inclusion/exclusion criteria and search methodology should be clearly documented.
3. **Dataset Analysis**
   Students should identify and compare the major datasets used in the field, considering:
   - dataset size;
   - number of subjects/users/documents;
   - labels and task definitions;
   - language and domain;
   - class distribution;

- availability and licensing;
- known limitations.

4. **NLP and Deep-Learning Methods**
   Organize previous research into meaningful methodological categories rather than simply presenting papers chronologically. For example:

Traditional ML → CNN/LSTM → BERT/Transformers → Domain-adapted Transformers → LLMs

The survey should explain what each approach contributes and where it performs well or poorly.

5. **Comparative Analysis**
   Students should construct tables comparing the literature across dimensions such as:

   Study Dataset Task Model Evaluation F1/AUROC Key Finding

6. Where appropriate, students should quantitatively summarize reported results, while being careful not to directly compare numbers obtained under incompatible experimental settings.

7. **Critical Analysis**
   This should be the **most important part of the paper**. Students should identify common weaknesses and methodological issues in the literature, such as:

   - small datasets;
   - patient/user-level data leakage;
   - inappropriate train/test splits;
   - lack of external validation;
   - inconsistent evaluation metrics;
   - demographic bias;
   - limited reproducibility;
   - lack of explainability;
   - poor generalization across datasets.

8. **Research Gap and Future Directions**
   The paper should conclude with **specific research opportunities**, rather than

simply stating that "more research is needed." Students should identify two or three concrete directions that follow from their analysis.

9. **Reproducible Survey Materials**
   Students should maintain a spreadsheet or structured database containing the papers reviewed and the information extracted from each paper. This can be submitted as supplementary material.

Recommended Scope

A semester survey should include approximately:

- **25–40 research papers**
- **5–10 major datasets**
- **3–6 methodological/model categories**
- **2–4 major research questions or comparison dimensions**
- **3–5 summary tables or figures**
- **6–8 pages of written content**, excluding references

Students should not attempt to survey an entire field such as *"NLP in healthcare."* The topic should be narrow enough that the literature can be analyzed critically within one semester.

What Makes the Survey a Research Paper?

The key distinction is between a **literature summary** and a **research survey**.

A weak survey would say:

"Researchers have used CNNs, LSTMs, BERT, and LLMs for dementia detection."

A stronger survey would ask:

"Do Transformer-based models actually improve dementia detection, or are reported improvements largely explained by differences in datasets and evaluation methodology?"

The students would then collect the evidence, organize the studies, compare experimental protocols, identify methodological problems, and determine what conclusions can reasonably be drawn.

Optional Quantitative Component

For stronger papers, students can conduct a **meta-analysis-style comparison** of published results. For example, they could investigate whether reported F1 scores differ systematically according to:

- model family;
- dataset size;
- dataset type;
- random vs. patient-level splitting;
- cross-validation vs. fixed test sets;
- AD vs. MCI vs. healthy-control classification.

They should not simply average scores from incompatible studies; instead, the analysis should explicitly account for differences in datasets and experimental protocols.

Suggested Semester Schedule

Weeks 1–2: Define topic and research questions; preliminary literature search
Weeks 3–4: Develop search strategy and inclusion/exclusion criteria
Weeks 5–7: Read and categorize papers; build literature database
Weeks 8–9: Analyze datasets and methodologies
Weeks 10–11: Comparative and critical analysis
Weeks 12–13: Identify research gaps and develop conclusions
Weeks 14–15: Write, revise, and present the paper

Expected Outcome

The final product should be a **focused, critical, and evidence-based survey**, rather than an encyclopedia-style review. A successful paper should allow the reader to understand **what has been done, which datasets and methods dominate the field, how reliable the reported findings are, what methodological limitations remain, and what research should be done next**.

For a particularly strong semester project, I would encourage students to frame the survey around a **controversial or unresolved question** rather than simply around a technology. That makes the resulting paper much more likely to have a meaningful contribution and potential for submission to a student research venue or NLP workshop.

A model paper: https://direct.mit.edu/tacl/article/doi/10.1162/TACL.a.35/133457/A-Systematic-Review-of-NLP-for-Dementia-Tasks